Faheem Nizar

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LinkedIn | Google Scholar

Education

Indian Institute of Technology (IIT) Kanpur Kanpur, India
Bachelor of Technology in Electrical Engineering; Minor in Machine Learning *Nov. 2020 – May 2024*

- **GPA**: **8.6/10** | Distinction for Academic Performance (2024); All India Rank 1136 in JEE Advanced 2020
- **Coursework**: Machine Learning, Big Data Visual Analytics, Image Processing, Computer Graphics, Data Structures & Algorithms, Probability & Statistics

Publications

Peer-reviewed Publications

E. Lumer, A. Gulati, **F. Nizar**, D. Hedroits, A. Mehta, H. Hwangbo, V. K. Subbiah, P. Basavaraju and J. Burke, "Tool and Agent Selection for Large Language Model Agents in Production: A Survey," *IEEE Conference on Artificial Intelligence*, May 2026.

F. Nizar, E. Lumer, A. Gulati, P. Basavaraju and V. K. Subbiah, "Agent-as-a-Graph: Knowledge Graph-Based Tool and Agent Retrieval for LLM Multi-Agent Systems," *ICAART*, Mar 2026.

S. Tripathi, **F. Nizar**, O. J. Pandey, T. Sandhan and R. M. Hegde, "ENADL: Towards Performance Improvement of IoT Networks Using Deep Learning-Based Node Fault Prediction," *IEEE Transactions on Reliability*, Mar 2025.

S. Dutta, **F. Nizar**, A. Amaan and A. Acharya, "Visual Analysis of Prediction Uncertainty in Neural Networks for Deep Image Synthesis," *IEEE Transactions on Visualization and Computer Graphics*, May 2024.

Preprints

E. Lumer, **F. Nizar**, A. Jangiti, K. Frank, A. Gulati, M. Phadate and V. K. Subbiah, "Don't Break the Cache: An Evaluation of Prompt Caching for Long-Horizon Agentic Tasks," arXiv preprint, Jan 2026.

E. Lumer, **F. Nizar**, A. Gulati, P. Basavaraju and V. K. Subbiah, "Tool-to-Agent Retrieval: Bridging Tools and Agents for Scalable LLM Multi-Agent Systems," arXiv preprint, Nov 2025.

Research Experience

Indian Institute of Technology Kanpur Kanpur, India
Undergraduate Researcher | Advisor: Prof. Tushar Sandhan *Jan 2024 – Jun 2024*

- Built high-fidelity simulations of large-scale IoT network communication in **NetworkX**, injecting physical node and link faults to reproduce real-world system failures
- Proposed an **Edge-and-Node Accumulation** graph deep learning architecture that learns spatio-topological signatures of faulty network behavior, reaching **98%** node-fault prediction accuracy
- Designed centrality-based resource allocation that extended network lifetime by **43%** through load redistribution

Indian Institute of Technology Kanpur Kanpur, India
Undergraduate Researcher | Advisor: Prof. Soumya Dutta *Jan 2023 – Mar 2023*

- Studied prediction uncertainty and confidence estimation in deep networks used for **deep image synthesis** on scientific datasets
- Compared **Ensemble** learning against **Monte Carlo Dropout** to quantify epistemic uncertainty, and identified the sample and ensemble counts needed for stable image generation
- Built visualization pipelines (parallel coordinate plots, 2D heatmaps) for multidimensional analysis of uncertainty metrics
- Found ensemble methods more robust than Monte Carlo Dropout for uncertainty quantification

Indian Institute of Technology Kanpur Kanpur, India
Research Intern, Antenna Tracker | Supervisor: Prof. Abhishek K. Gupta *May 2022 – Aug 2022*

- Built a real-time antenna tracking system to hold long-range, low-latency links between UAVs and a ground station
- Set up **MAVLink** bidirectional telemetry over ESP-NOW on ESP32, and a self-healing **IEEE 802.11s** BATMAN mesh on Raspberry Pi nodes for swarm UAV communication
- Engineered **PWM-to-PPM** conversion to drive stepper motors from a Pixhawk flight controller for precise actuation
- Integrated a **YOLOv5** visual tracking module to detect and follow UAVs in real time for closed-loop antenna alignment for short ranges

Professional Experience

PwC US Bangalore, India
Senior Associate, Data Scientist *Jul 2024 – Present*

- Developed an agentic red-teaming framework for physical-safety evaluation of robot foundation models, combining **LLM-guided scenario generation** and **MAP-Elites** search over object placement, latent physics, and environmental disturbances to identify diverse failure modes across four policies
- Collected **100+** teleoperated WidowX manipulation episodes with synchronized dual-camera observations and robot states, fine-tuned $\pi_{0.5}$ model using **JAX/LoRA** for 20-step action prediction and built a **MuJoCo** environment for policy testing
- Helped establish PwC's **Physical AI Lab**, integrating the **WidowX AI Follower** robotic arm and GPU infrastructure to support manipulation experiments and evaluation of VLA policies
- Researched **graph-based tool and agent retrieval** for LLM systems with hundreds of available tools, contributing to research publications and retrieval methods subsequently integrated into production systems
- Built a **text-conditioned 3D motion generation** system for synthesizing character animations from natural-language prompts, reducing motion-capture requirements for NPC animation
- Trained and deployed a **DistilBERT** classifier using **Databricks** and **MLflow**, reducing a benchmarking workflow from **4 weeks to 10 minutes**

American Express Bangalore, India
Data Analyst Intern (Pre-Placement Offer) *May 2023 – Jul 2023*

- Built a credit-default risk framework for low-tenure, high-balance cardholders by engineering risk variables and training an **XGBoost** model on internal and external tradeline data
- Ran large-scale segment simulations for a **2%** GINI gain and **4%** higher default capture, with 3 of 4 engineered features in the top 100 by model importance

Selected Projects & Competitions

GenCast Hurricane Forecasting | *FutureSprint Hackathon* Oct 2025

- Ran the **GenCast** mini ensemble model to predict **15-day** hurricane track ensembles from ECMWF weather data
- Mapped predicted tracks to affected zip codes for targeted warnings, and generated stakeholder reports using a grounded web-search agent

IoT Innovation Contest | *FutureThink Cloud Innovation Summit* Sep 2024

- Integrated agentic AI into IoT kiosks using Raspberry Pi, OpenCV face detection, and a conversational assistant

Technical Skills

Languages: Python, C++, C, SQL
ML / DL: PyTorch, PySpark, Hugging Face Transformers, XGBoost, MLflow, AWS Sagemaker
Vision & graphs: OpenCV, YOLO, NetworkX
Systems & tools: Linux, Docker, Databricks, Git, ROS2, MuJoCo

Leadership & Service

PwC US Bangalore, India
Learning & Development Lead, Data Science Team *Jul 2024 – Present*

- Ran internal sessions on prompt engineering and agentic AI, and managed staffing across a **46-member** team

Academics and Career Council, IIT Kanpur Kanpur, India
Head of Web Development *May 2022 – May 2023*

- Led a team of **6** developers building portals used across the campus community